SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE, AMROLI
MSc IT PROGRAMME
MODEL QUESTION PAPER - 2

Course: FY MSc IT / BSc IT	**Year: 2026-27**
Semester: 1st	**Total Marks : 50**
Subject: 104 Fundamentals of Programming Using C-1	**Time: 2 Hours**

Q.1	**Do as directed: (Attempt any Ten)**	**[10]**
1	What is the full form of ASCII?	
2	Which header file must be included to use standard input and output functions like printf() and scanf()?	
3	What is the result of 20 / 3 in integer arithmetic in C?	
4	In a flowchart, which geometric symbol is used to represent decision or condition checking?	
5	The % (modulus) operator can only be applied to which data types? (A) float (B) int (C) double (D) all data types	
6	True or False: In C, if(0) condition evaluates to False.	
7	Which statement is used to skip the rest of the current loop iteration and jump to the next iteration?	
8	Name the three levels of computer programming languages.	
9	What is the format specifier for reading a double variable using scanf()?	
10	Which of the following is an invalid variable name? (A) student_age (B) _roll (C) float (D) marks1	

11	What will be the output of the following code snippet?
	int a = 10;
	if(a = 5)
	printf("True");
	else
	printf("False");	

12	Write the escape sequence used for a horizontal tab in C.	
Q.2 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	Explain the primary data types available in C (int, float, char, double) along with their qualifiers (short, long, signed, unsigned). Explain the steps/phases of Problem-Solving Methodology in programming with an illustrative diagram. Explain while loop and do...while loop with syntax, flowcharts, and compare their key differences.	

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Q.2 (B)	**Answer following question: (Attempt any One)**	**[04]**
(1) **OR** (2)	Explain Relational and Logical operators in C with truth tables and code snippets. Explain the role of Assembler, Compiler, and Linker in the program development life cycle.	
Q.3 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	Define Algorithm. What are the key characteristics of a good algorithm? Write an algorithm to find whether a number is positive, negative, or zero. Explain if...else ladder and nested if statements with syntax and a program to assign grades based on student marks. What is Program Verification? Explain different types of bugs (syntax, runtime, logical) and how testing is performed.	
Q.3 (B)	**Answer following question: (Attempt any One)**	**[04]**
(1) **OR** (2)	Explain Type Casting in C. Differentiate between Implicit and Explicit type conversion with examples. Explain break and goto statements in C. Why is the excessive use of goto discouraged?	
Q.4	**Answer following questions: (Attempt any Three)**	**[12]**
(1) (2) (3) (4)	Write a C program to check whether a given character entered by the user is a Vowel or a Consonant. Draw a flowchart and write an algorithm to find the area and circumference of a circle for a given radius R. Write a C program to calculate the sum of first N natural numbers (1 + 2 + ... + N) where N is entered by the user. Write a C program to check whether a given integer number is a Palindrome (e.g., 121 is a palindrome).	